PROJECT AND TEAM INFORMATION

Project Title

Cyber Case Management System

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Motivation for Cyber Security Management System
In today's digital world, technology has become an essential part of almost every aspect of life. Individuals, businesses, educational institutions, banks, hospitals, and government organizations depend heavily on computers, networks, cloud platforms, and online services to store and exchange information. While this digital transformation has made communication and business processes faster and more convenient, it has also increased the risk of cyber threats. Cyberattacks such as phishing, malware, ransomware, password attacks, identity theft, data breaches, and unauthorized access are becoming increasingly common and sophisticated.
The motivation for a Cyber Security Management System comes from the need to protect digital assets and sensitive information from these growing threats. A single security incident can result in financial losses, privacy violations, loss of important data, and damage to an organization's reputation. Therefore, cybersecurity cannot be limited to installing antivirus software or using strong passwords; it requires continuous monitoring, proper risk assessment, access management, incident handling, and security awareness.
A Cyber Security Management System provides a structured approach to identifying and managing security risks. It helps organizations understand their vulnerabilities, monitor suspicious activities, control access to resources, and respond effectively when security incidents occur. It can also help maintain records of incidents, security policies, and preventive measures, making it easier to improve security over time.

State of the Art / Current solution (1 pt)

Today, cybersecurity problems are solved using a combination of technologies, policies, and human awareness. Organizations use firewalls and antivirus software to block malicious activities and detect malware. Encryption protects sensitive data while it is stored or transferred. Multi-factor authentication adds an extra layer of security to user accounts. Intrusion detection and monitoring systems identify unusual network activity and alert security teams. Regular software updates and security patches fix known vulnerabilities. Organizations also conduct backups to recover data after ransomware or other attacks. Employee training helps users recognize phishing emails and unsafe links. Together, these measures reduce cyber risks and protect digital systems from evolving threats.

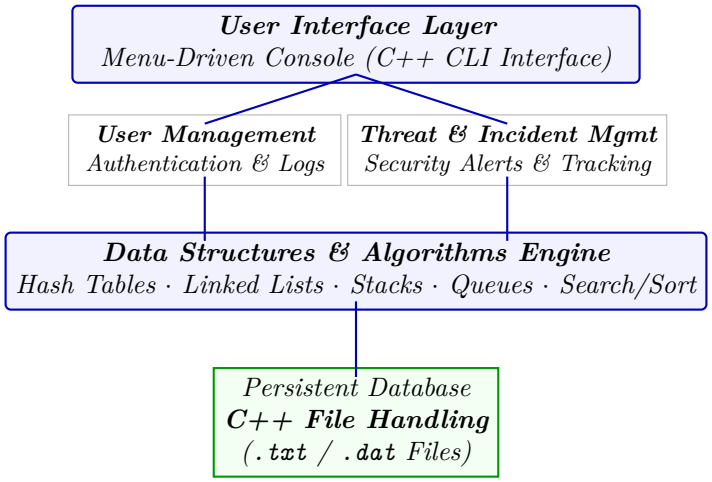
Project Goals and Milestones (2 pts)

The general goal of a Cyber Security Management System is to provide an affordable and effective solution for protecting digital systems, networks, and sensitive information from common cyber threats. The system aims to offer essential cybersecurity features without requiring expensive hardware, software, or specialized infrastructure. It focuses on identifying potential security risks, monitoring suspicious activities, managing user access, protecting important data, and providing basic incident detection and response capabilities. Another important goal is to make cybersecurity accessible to small organizations, educational institutions, startups, and individual users who may have limited budgets and technical resources. The system also aims to improve security awareness by encouraging users to follow safe practices such as strong authentication, secure password management, and protection against phishing and unauthorized access. By combining essential security functions into a simple and cost-effective platform, the system can help reduce the possibility of data loss, unauthorized access, and cyberattacks. Overall, the goal is to develop a reliable, user-friendly, scalable, and economical cybersecurity management solution that provides adequate protection while keeping implementation and maintenance costs low.

Project Approach (3 pts)

The Cyber Security Management System will be developed as a C++-based application using fundamental data structures and algorithms. The main objective is to manage cybersecurity-related information such as users, login records, security alerts, threats, and incidents in an organized and efficient manner. The project will be developed using C++ in an IDE such as Visual Studio Code, Code::Blocks, or Dev-C++. Different data structures will be selected according to the requirements of each module. Arrays and structures can be used to store basic user and security information, while linked lists can be used to maintain dynamic records of security incidents. Stacks can be used for maintaining recent actions or supporting undo-type operations, whereas queues can be used to manage security alerts or incidents in the order they are received. Searching and sorting algorithms will be implemented to quickly find users, threats, or incident records and arrange them based on factors such as priority or severity. A hash table can also be used for faster user or password-related record searching. File handling in C++ can be used to store and retrieve records, allowing the system to maintain information even after the program is closed. The project will have a simple menu-driven console interface, making it easy to operate and suitable for a second-year B.Tech level. The system will be tested using sample cybersecurity scenarios to verify the correctness and efficiency of the implemented data structures and algorithms. Overall, the approach combines C++ programming, data structures, algorithms, and basic cybersecurity concepts to create a simple, practical, and low-cost security management system.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The main outcome of the Cyber Security Management System is a functional C++-based application that demonstrates the practical use of data structures and algorithms for managing cybersecurity-related information. The system will provide a menu-driven interface for managing users, security threats, alerts, incidents, and security records. Different data structures such as arrays, linked lists, stacks, queues, and hash tables will be implemented according to the requirements of different modules. Searching and sorting algorithms will be used to efficiently locate and organize records. C++ file handling will be used to store and retrieve user details, threat information, incident records, and security logs. The project will also demonstrate basic concepts of authentication, threat identification, incident tracking, and security monitoring. The final deliverables will include the working C++ application, implemented data structures and algorithms, file-handling functionality, sample test cases and outputs, system architecture, algorithm descriptions, project documentation, and a final report. Overall, the project will demonstrate how C++ programming and data structures can be applied to develop a practical cybersecurity management system.

Assumptions

For this cybersecurity project, we assume that users will follow basic security guidelines and provide accurate information when required. We assume that the system will have reliable internet access, sufficient computing resources, and access to updated security tools. We also assume that the system can detect common cyber threats and suspicious activities and generate alerts in time. Important data is assumed to be securely stored, backed up, and recoverable when necessary. We assume that authorized users will be allowed access while unauthorized users will be blocked. Finally, we assume that the project will focus on common cybersecurity threats and can be updated to handle new threats in the future.

References

M. T. Goodrich, R. Tamassia, and M. H. Goldwasser, Data Structures and Algorithms in C++, Wiley.
Robert Lafore, Data Structures and Algorithms in Java, Sams Publishing. (Useful for understanding data-structure concepts, even though the implementation is in C++.)
Bjarne Stroustrup, Programming: Principles and Practice Using C++, Addison-Wesley.
GeeksforGeeks – Data Structures and Algorithms
cppreference – C++ Language and Standard Library Reference
W3Schools – C++ Tutorial
GitHub – Code Repository and Version Control Platform
Microsoft Visual Studio Code – Development Environment
OWASP – Cybersecurity and Application Security Resources
NIST Cybersecurity Framework